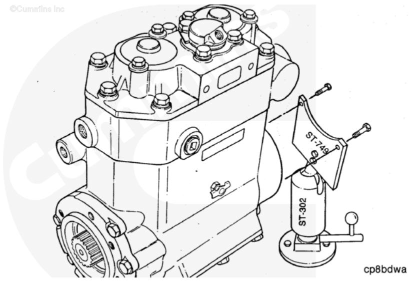
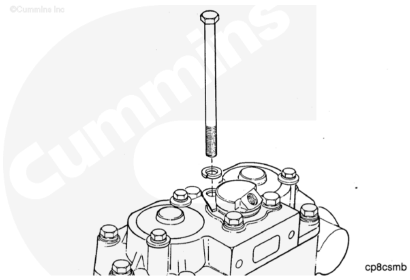
Trainings

Disassemble

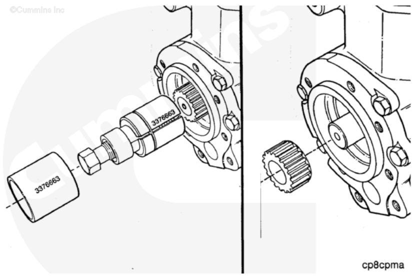
Install the air compressor mounting plate, Part Number ST-749, or equivalent, which is used with the ball joint vise, Part Number ST-302, or equivalent.



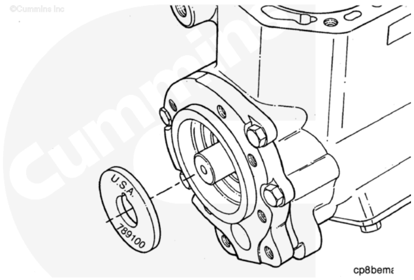
Disassemble the air compressor cylinder head. Refer to Procedure 012-106 in Section 12 (/qs3/pubsys2/xml/en/procedures/04/04-012-106.html).



Use a coupling puller, Part Number 3376663, or equivalent, to remove the spline coupling hub.

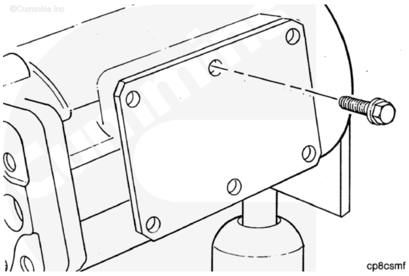


Remove the front thrust bearing.

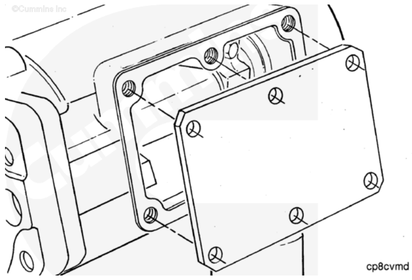


Remove the three flange head capscrews.

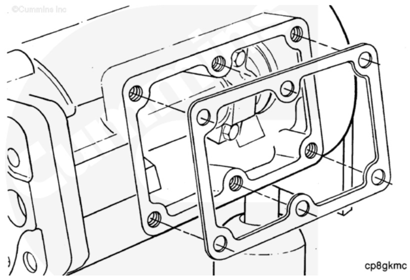
Remove the three captive washer capscrews.



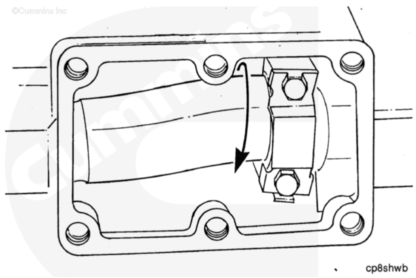
Remove the access cover.



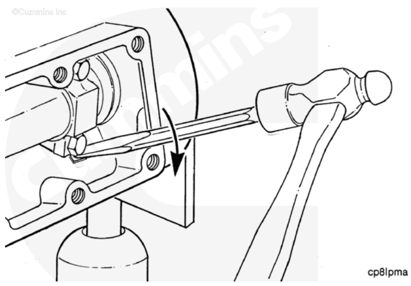
Remove and discard the access cover gasket.



Rotate the crankshaft so that one piston is at bottom dead center.

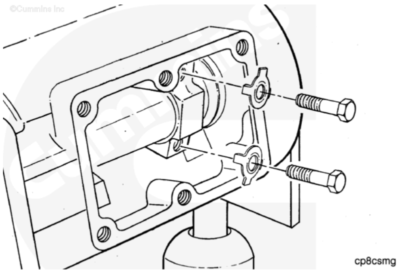


Use a chisel and hammer to move the lock plate tangs away from the hexagon head capscrew heads.



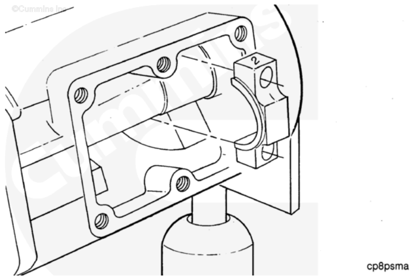
Remove the two hexagon head capscrews and the two lock plates.

Discard the lock plates.



Note : Make sure the cap is numbered on the crankcase dataplate side. If not, do so at this time.

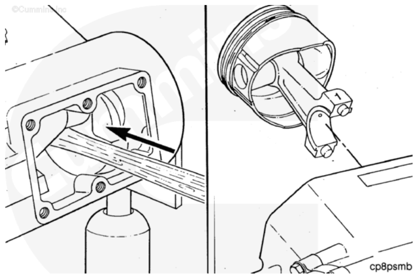
Remove the connecting rod cap.



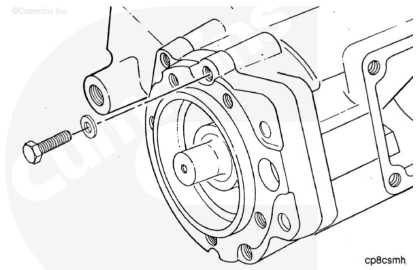
Note : Make sure the cap is numbered on the crankcase dataplate side. If not, do so at this time.

Use a hammer handle to push the piston and connecting rod assembly out of the crankcase.

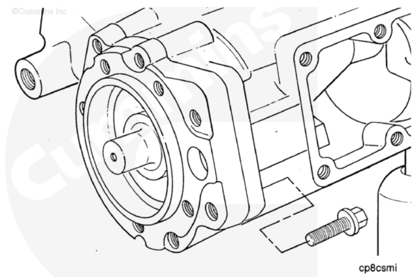
Repeat the last five steps to remove the other piston and connecting rod assembly.



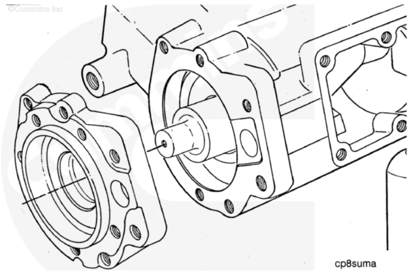
Remove the four twelve point capscrews and four lock washers.



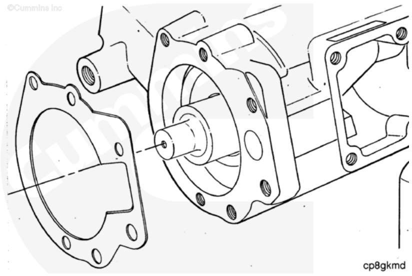
Remove the two captive washer capscrews.



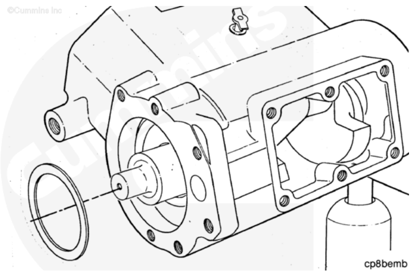
Remove the air compressor support.



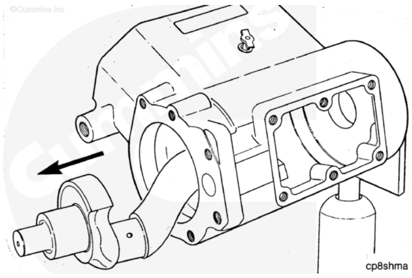
Remove and discard the air compressor support gasket.



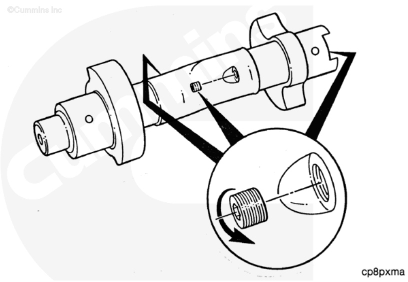
Remove the rear thrust bearing.



Remove the crankshaft.



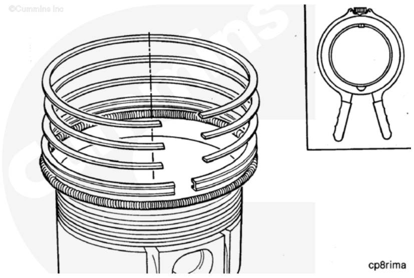
Remove the three pipe plugs.



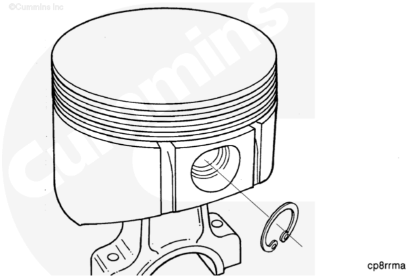
Install the piston and connecting rod assembly in the soft-jawed vise.

If the rings are to be used again, tag each one as to the position from which it was removed.

Remove the top and second compression rings, and the three-piece oil ring.



Remove the two retaining rings.

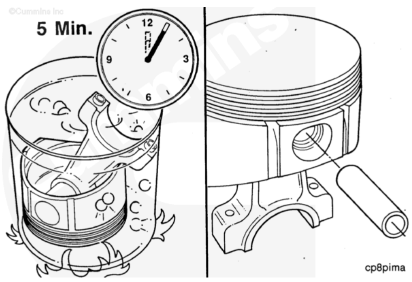


⚠ WARNING ⚠

Improper practices, carelessness, or ignoring the warnings can cause burns, cuts, mutilation, asphyxiation, or other personal injury or death.

⚠ CAUTION ⚠

Do not force the piston pin from the piston or the piston can be damaged.

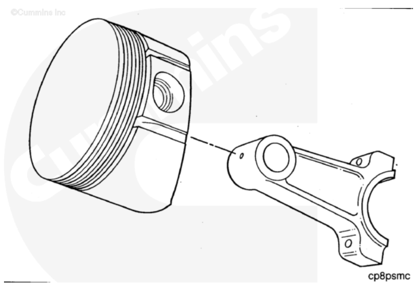


If the pin can **not** be removed by hand pressure, place the piston in boiling water for five minutes to expand the pin bore. This will allow the pin to be removed.

Remove the piston pin.

Remove the piston.

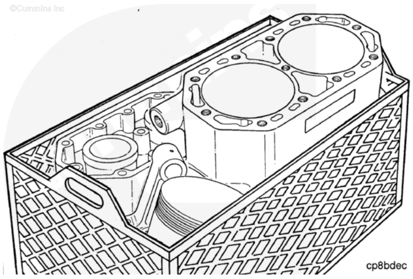
Repeat the last six steps to disassemble the other piston and connecting rod assembly.



Clean and Inspect for Reuse

⚠ WARNING ⚠

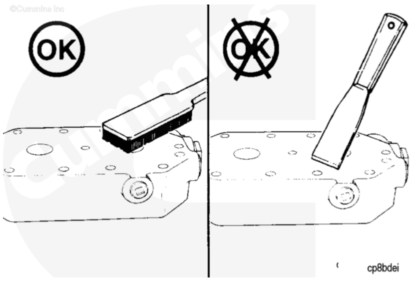
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Soak the parts in a kerosene emulsion based cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier can be contacted about solution concentration, temperature, or soak time.

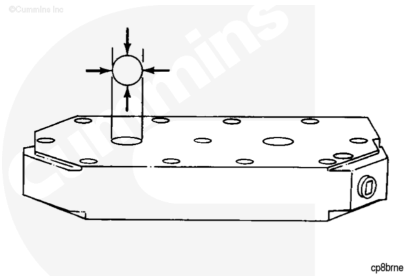
⚠ CAUTION ⚠

Do not use a scraper to remove carbon and scale or the sealing surfaces can be damaged.

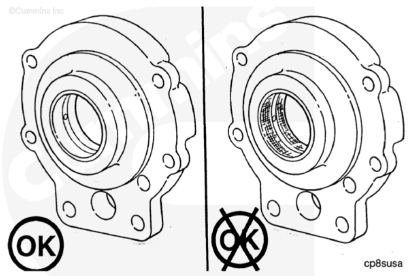


The parts can be scrubbed with a stiff non-metallic bristle brush.

Inspect the cylinder head and components. Refer to Procedure 012-106 in Section 12 (/qs3/pubsys2/xml/en/procedures/04/04-012-106.html).

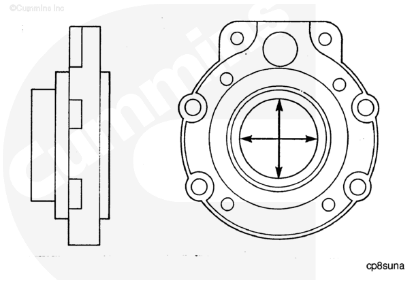


Inspect the bushing in the air compressor support. Replace if damaged.



Measure the inside diameter.

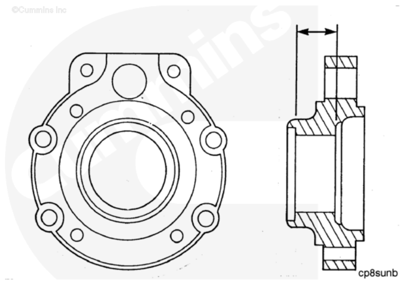
Replace the bushing if worn larger than 47.66 m [1.877 in].



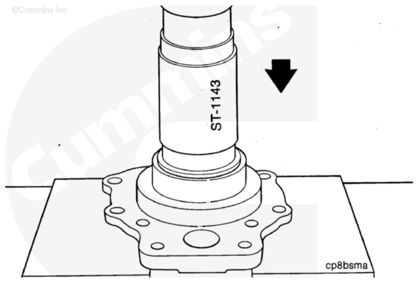
Measure the dimension between thrust faces.

Replace the support if worn beyond the following dimension:

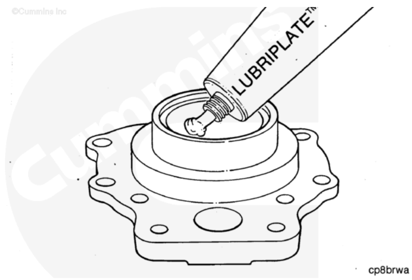
Thrust Face Dimension			
	mm		in
199351	30.84	MIN	1.214
	31.00	MAX	1.220
204056	30.84	MIN	1.214
	31.00	MAX	1.220



Use an air compressor bushing mandrel, Part Number ST-1143, or equivalent, to remove the worn bushing.

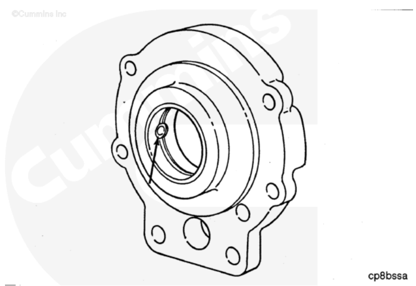


Use clean Lubriplate™ to lubricate the bushing bore in the air compressor support.



⚠ CAUTION ⚠

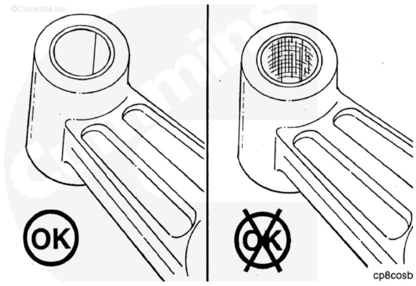
Make sure the oil holes in the bushing and the support are aligned. Failure to align the oil holes can result in bushing failure.



Use an air compressor bushing mandrel, Part Number ST-1143, or equivalent, to install the new bushing.

Push in the new bushing until it is flush with the bore surface.

Inspect the connecting rod piston pin bore for damage.
Replace if damaged.

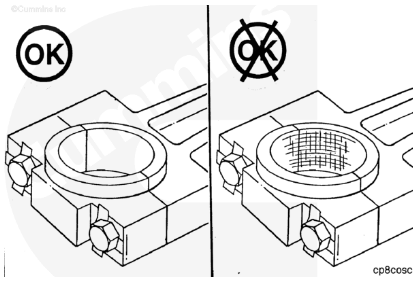


Install new lock plates and the hexagon head capscrews.
Tighten the capscrews.

Torque Value: 23 n•m [16 ft-lb]

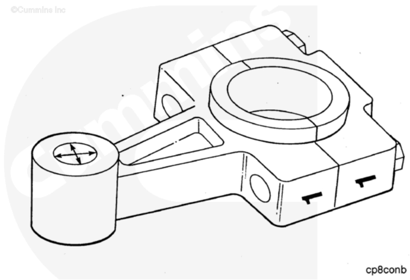
Measure the inside diameter of the crankshaft end of the connecting rod.

If the inside diameter exceeds 39.79 mm [1.567 in], replace the connecting rod.



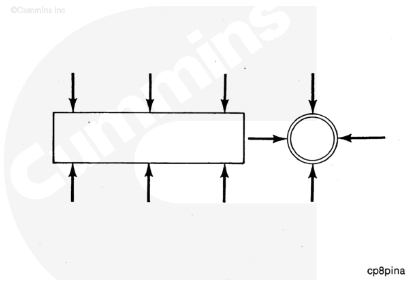
Measure the inside diameter of the piston end of the connecting rod.

If the inside diameter exceeds 17.50 mm [0.689 in], replace the connecting rod.



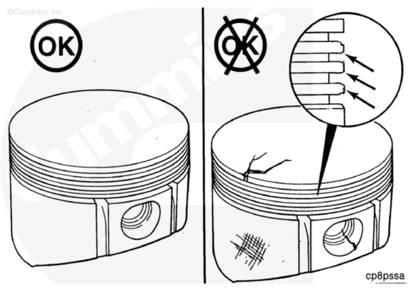
Measure the piston pin outside diameter.

Replace if worn smaller than 17.45 mm [0.687 in].



Inspect the piston top and pin bore for cracks.

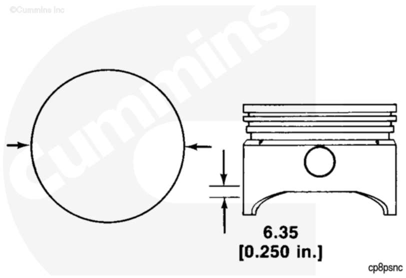
Check the ring grooves and skirt for damage. Replace if damaged.



Measure the outside diameter of the piston.

Measure at 6.35 mm [0.250 in] above the bottom of the piston skirt and at a right angle to the piston pin bore.

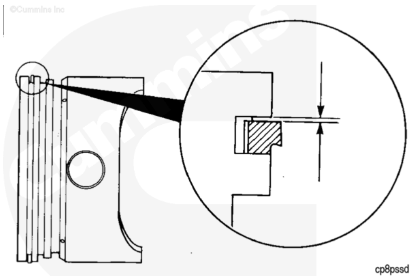
Replace if worn smaller than 91.86 mm [3.617 in].



Use new piston rings and feeler gauge to measure for ring groove wear.

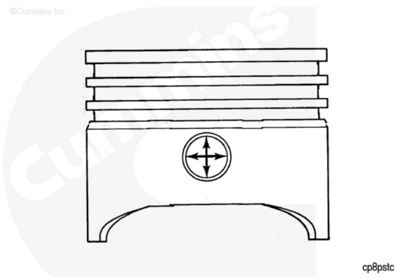
Replace the piston if the gaps are worn larger that the following dimensions:

Ring to Groove Clearance			
	mm		in
Piston Ring (top)	0.05	MIN	0.002
	0.114	MAX	0.0045
Piston Ring (second)	0.05	MIN	0.002
	0.114	MAX	0.0045
Piston Ring (oil)	0.038	MIN	0.0015
	0.10	MAX	0.004



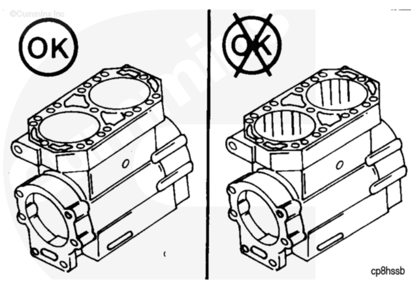
Measure the piston pin bore.

Replace the piston if the bore is worn larger than 17.49 mm [0.689 in].



Inspect the crankcase cylinder bores for scoring or other damage.

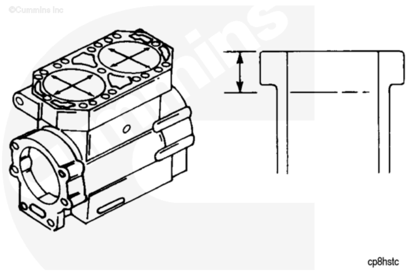
If the cylinder bores are scored or damaged, bore or hone the cylinder for oversized piston and rings.



Use a dial bore gauge, Part Number 3375072, or equivalent, to measure the cylinder bore diameter.

Measure at 25.00 mm [1.00 in] below the top of the crankcase.

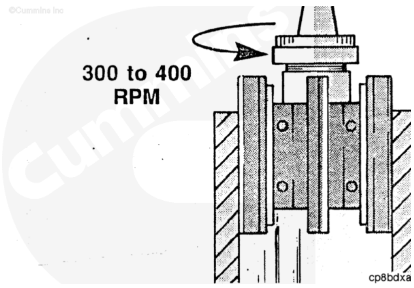
Maximum out-of-round is 0.04 mm [0.002 in].



Air Compressor Cylinder Bore		
mm		in
92.08	MIN	3.625
92.164	MAX	3.6285

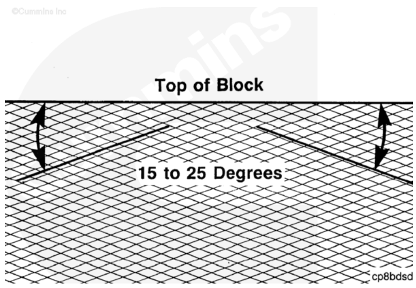
If the crankcase bore is **not** within specification bore the cylinder to use oversize piston and rings.

Bore or hone the cylinders (per the following instructions) to the following specifications:



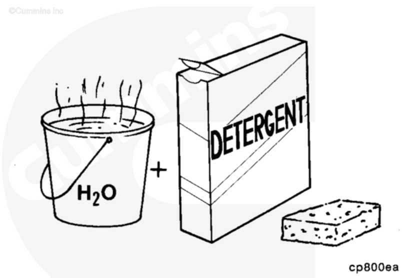
Air Compressor Cylinder Bore	
Oversize mm [in]	**Diameter mm [in]**
0.25 [0.010]	92.35 [3.636]
0.51 [0.020]	92.61 [3.646]
0.76 [0.030]	92.86 [3.656]

Use a 280 grit hone to produce a cross hatched surface finish with lines at a 15 to 25 degree angle with the top of the crankcase.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

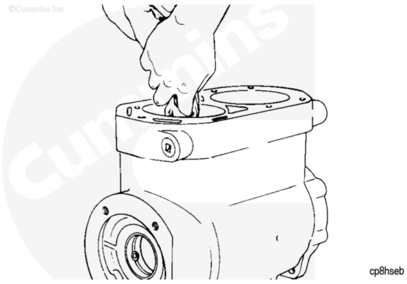


Clean the cylinder bores with a strong solution of laundry detergent and hot water.

Dry with compressed air.

Check the bores for cleanliness by wiping with a white, lint-free, lightly oiled cloth.

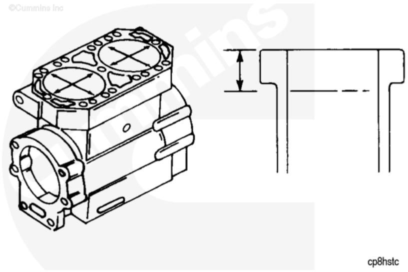
If grit is still present, clean again.



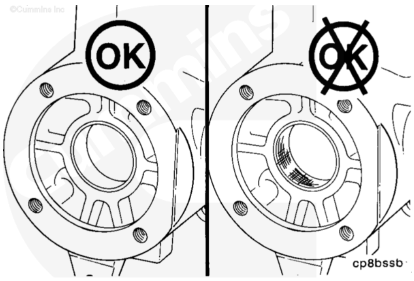
Use a dial bore gauge, Part Number 3375072, to measure the cylinder bore diameter.

Measure at 25.00 mm [1.00 in] below the top of the crankcase.

Maximum out-of-round is 0.04 mm [0.002 in].

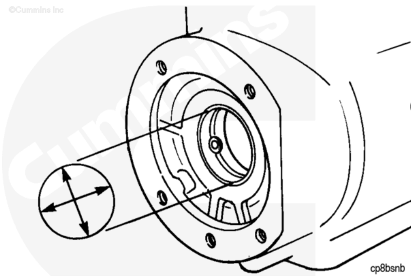


Inspect the crankcase bushing in the crankcase. Replace, if damaged.

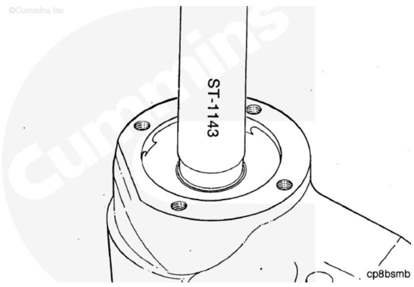


Measure the inside diameter.

Replace the air compressor crankcase bushing if worn larger than 47.7393 mm [1.8795 in].

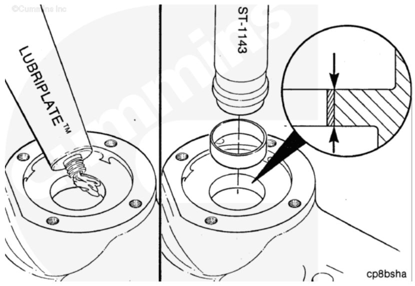


Use an air compressor bushing mandrel, Part Number ST-1143, or equivalent, to remove the bushing.



⚠ CAUTION ⚠

Make sure the oil holes in the bushing and crankcase are aligned. Failure to do so can result in bushing failure.

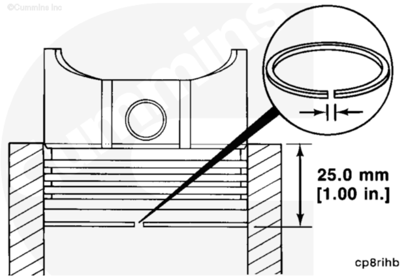


Use clean Lubriplate™ to lubricate the bushing bore in the crankcase.

Use an air compressor bushing mandrel, Part Number ST-1143, or equivalent, to install the new bushing.

Push in the new bushing until it is flush with the bore surface.

Note : Cummins Inc. recommends that new piston rings be installed during rebuild.



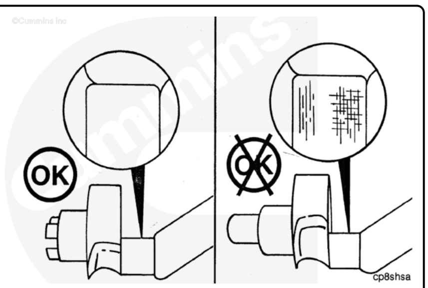
If old rings are used, follow the following instructions to measure the ring gaps.

Insert one ring at a time into the cylinder bore. Seat the ring with a piston head squarely 25.00 mm [1.00 in] below the top of the crankcase.

If the ring gaps exceed the specification listed below, the rings **must** be replaced.

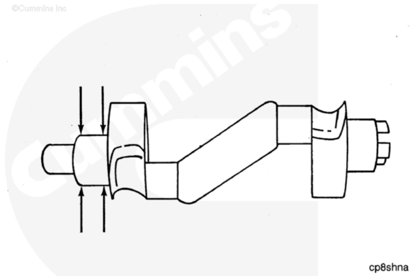
Piston Ring Gaps			
	mm		in
Piston Ring (top)	0.25	MIN	0.010
	0.51	MAX	0.020
Piston Ring (second)	0.25	MIN	0.010
	0.51	MAX	0.020
Piston Ring (oil)	0.38	MIN	0.140
	0.015	MAX	0.055

Inspect the crankshaft for scratches and scoring. Replace, if damaged.



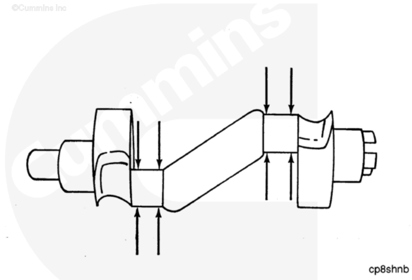
Measure the outside diameter of the journal at the front air compressor support.

Replace if worn beyond specifications given in the Crankshaft Dimensions Chart following the next two steps.



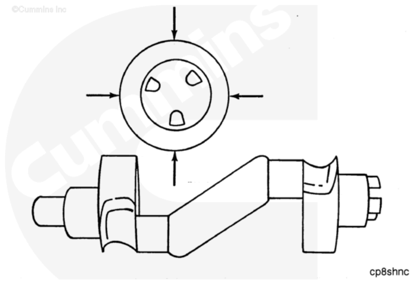
Measure the outside diameter of the journal at the connecting rods.

Replace if worn beyond specifications in the Crankshaft Dimensions chart following the next step.



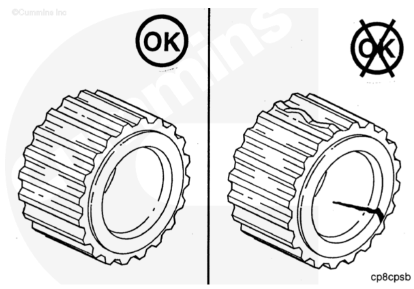
Measure the outside diameter of the journal at the rear coupling.

Replace if worn beyond specifications in the following Crankshaft Dimension chart.



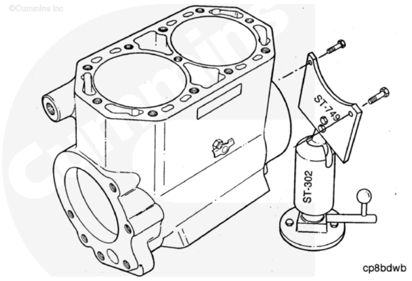
Crankshaft Dimensions						
	Front Support Journal		Connecting Rod Journal		Rear Coupling Journal	
Crankshaft Part Number	Min mm [in]	Max mm [in]	Min mm [in]	Max mm [in]	Min mm [in]	Max mm [in]
200370	47.52 [1.871]	47.57 [1.873]	39.62 [1.560]	39.69 [1.563]	47.52 [1.871]	47.57 [1.873]

Inspect the spline coupling hub for cracks and worn or broken teeth. Replace, if damaged.



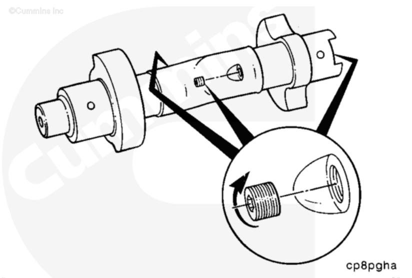
Assemble

Install the crankcase to the mounting plate, Part Number ST-749, or equivalent, which is used with the ball joint vise, Part Number ST-302, or equivalent.



⚠ CAUTION ⚠

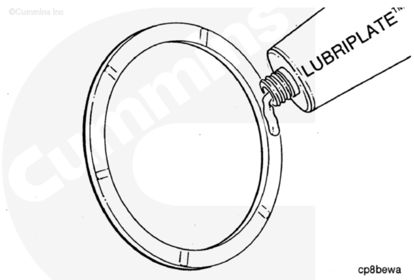
Omission of these plugs can result in excessive oil carryover into the discharge air stream causing contamination in air compressor air lines.



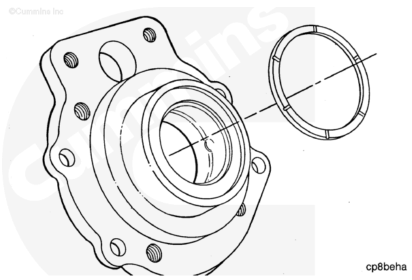
Install the three pipe plugs. Tighten the pipe plugs.

Torque Value: 7 n•m [62 in-lb]

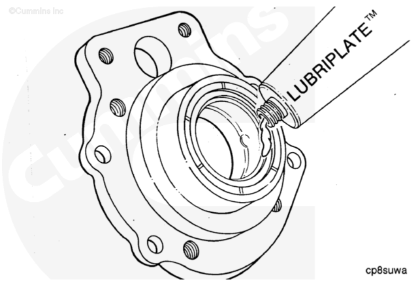
Use clean Lubriplate™ to lubricate the rear thrust bearings.



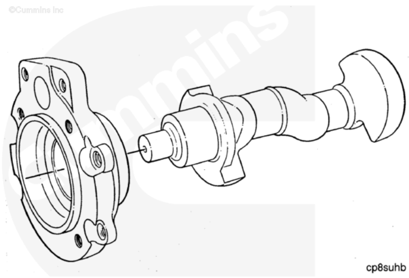
Install the rear thrust bearing on the support seating surface with the grooved side away from the support.



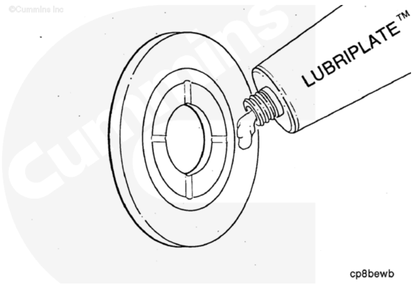
Use clean Lubriplate™ to lubricate the support bushing bore.



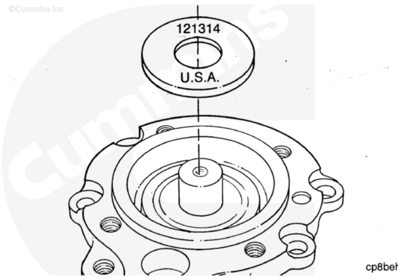
Install the crankshaft into the air compressor support.



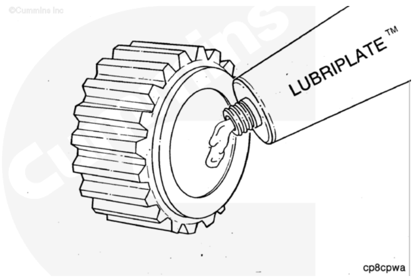
Use clean Lubriplate™ to lubricate the front thrust bearing.



Install the front thrust bearing on the crankshaft with the grooved side toward the air compressor support.

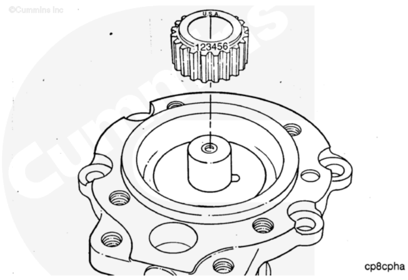


Use clean Lubriplate™ to lubricate the spline coupling hub bore.



Note : Support the crankshaft while pressing on the spline coupling hub.

Use a press to install the spline coupling hub with the part number side or side marked "OUT" on the crankshaft.

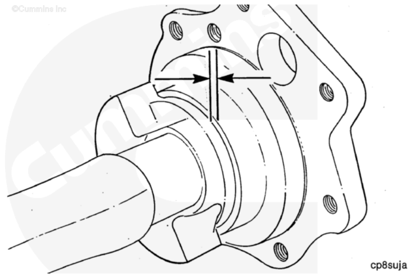


Measure the crankshaft to support clearance.

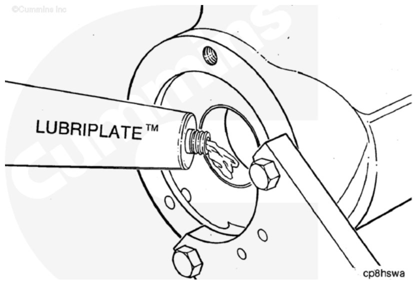
Clearance **must** be 0.038 to 0.191 mm [0.0015 to 0.0075 in].

To obtain proper clearance, reference the following specifications:

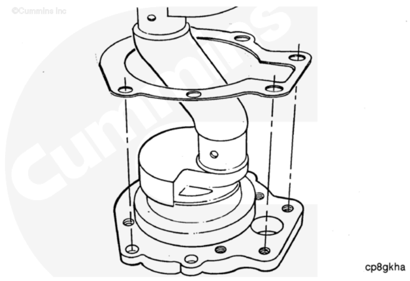
Thrust Bearing Size in Support Flange			
Thrust Bearing		mm	in
Front	211662	6.10 to 6.30	0.240 to 0.248
Rear	188040	2.27 to 2.30	0.090 to 0.091
188042		2.30 to 2.32	0.091 to 0.092
188044		2.32 to 2.35	0.092 to 0.093



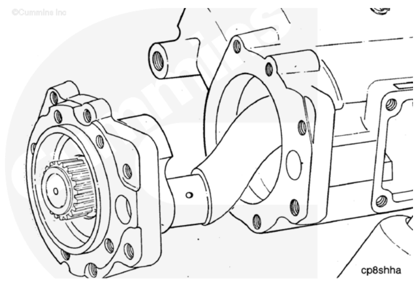
Use clean Lubriplate™ to lubricate the crankcase bushing bore.



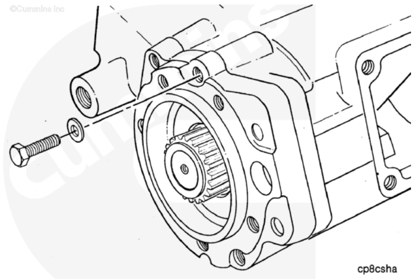
Install a new support gasket to crankcase.



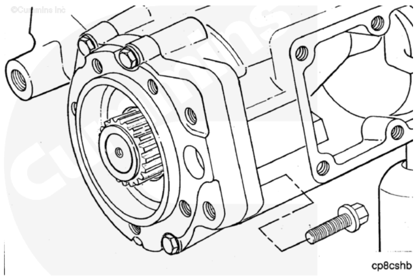
Carefully install the crankshaft and support assembly into the crankcase.



Install the four lock washers and the four twelve point capscrews.

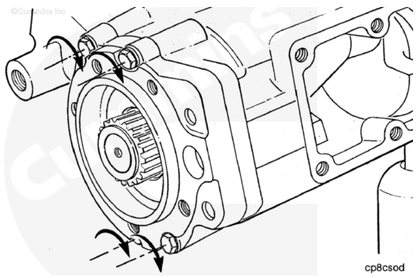


Install the two hexagon head capscrews.

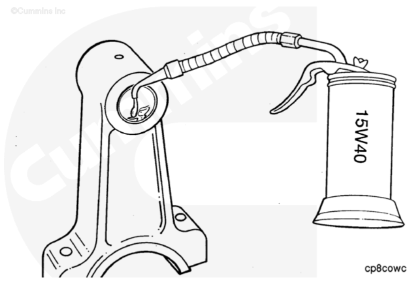


Tighten the capscrews.

Torque Value: 47 n•m [35 ft-lb]

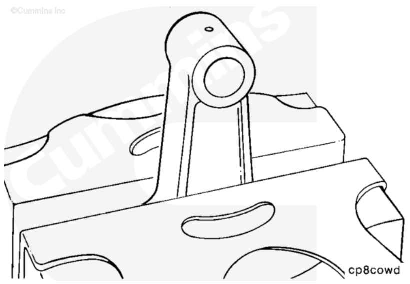


Use clean 15W-40 lubricating engine oil to lubricate the piston pin bore of the connecting rod.

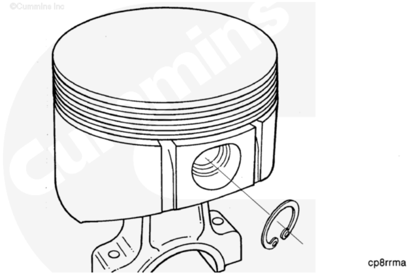


Note : If new rods are being installed, stamp 1 or 2 on the rod and cap for future reference.

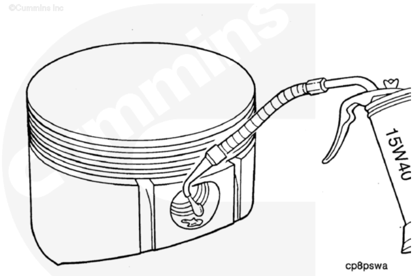
Install the connecting rod in the soft-jawed vice.



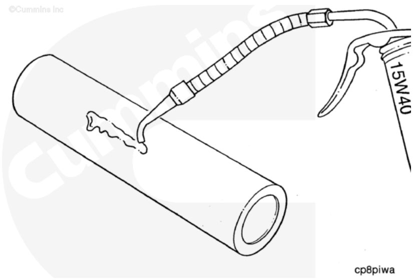
Install on piston pin retaining ring.



Use clean 15W-40 lubricating engine oil to lubricate the piston pin bore of the piston.



Use clean 15W-40 lubricating engine oil to lubricate the piston pin outside diameter surface .

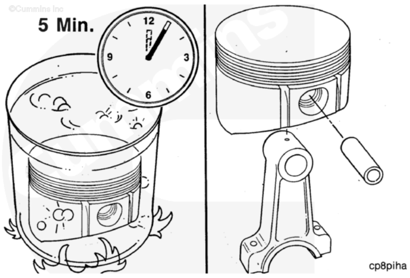


⚠ WARNING ⚠

Improper practices, carelessness, or ignoring the warnings can cause burns, cuts, mutilation, asphyxiation, or other personal injury or death.

⚠ CAUTION ⚠

Do not force the piston pin from the piston or the piston can be damaged.

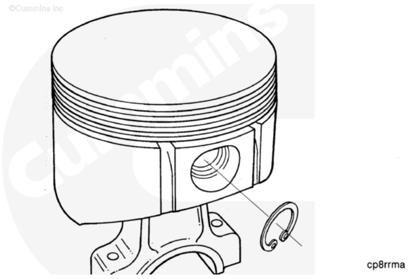


If the piston pin can **not** be installed by hand pressure, heat the piston in boiling water for 5 minutes to expand the pin bore. This will enable the piston pin to be installed.

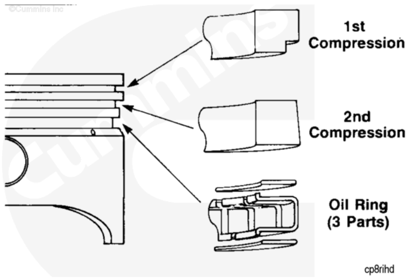
Use clean 15W-40 lubricating engine oil to lubricate the piston pin bore.

Install the piston pin.

Install the second piston pin retaining ring.



Position the piston rings.



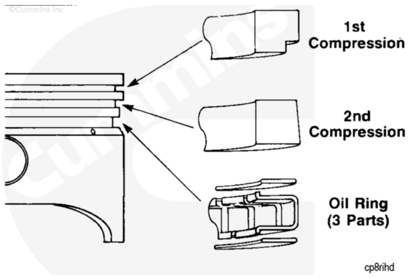
Install the bottom oil ring rail below the piston oil ring groove.

Install the oil ring expander in the piston oil ring groove. Move the bottom oil ring rail below the oil ring expander.

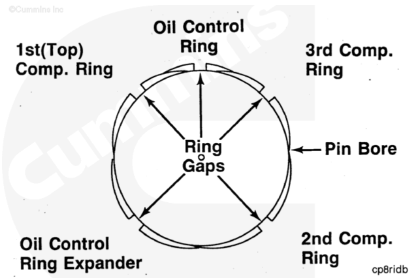
Install the top oil ring rail in the piston oil ring groove above the oil ring expander.

Use a piston ring expander to install the second piston ring, part number side or side marked “TOP” up.

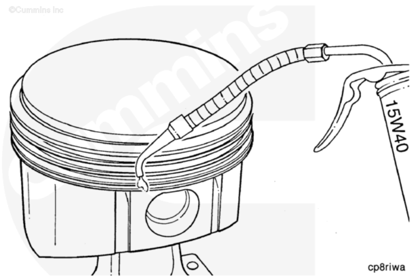
Use a piston ring expander to install the top piston ring, part number side or side marked “TOP” up.



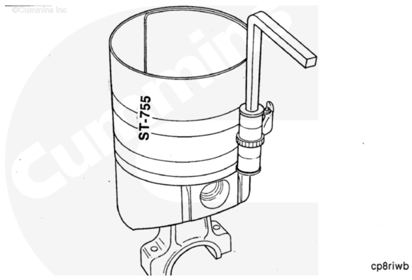
Position the piston ring gaps so they are **not** over the piston pin bores, and separate the oil ring rails 45 to 90 degrees.



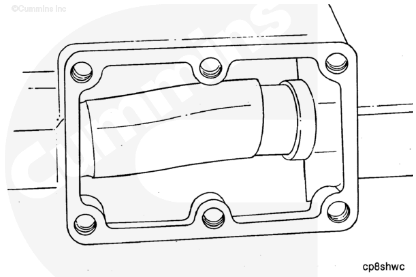
Use clean 15W-40 lubricating engine oil to lubricate the piston rings.



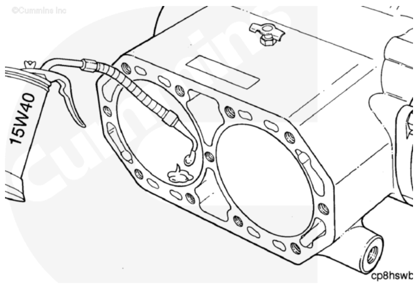
Use a piston ring compressor, Part Number ST-755, or equivalent, to compress the piston rings.



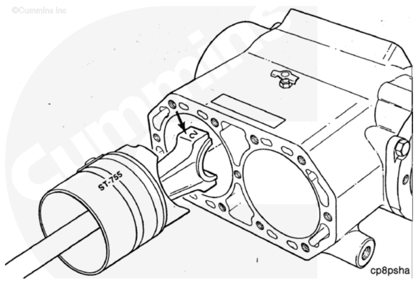
Rotate the crankshaft so that one of the connecting rod journals is at bottom dead center.



Use clean 15W-40 lubricating engine oil to lubricate the crankcase piston bore.

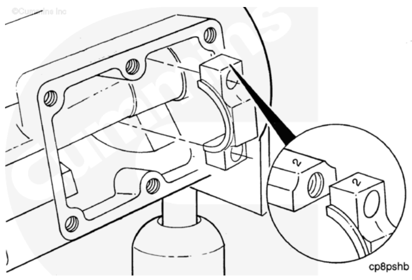


Install the piston and connecting rod into the cylinder bore with the numbered side of the connecting rod toward the dataplate side of the crankcase.

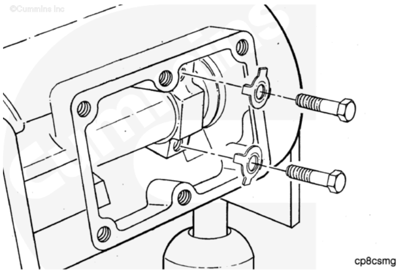


Install the matching cap to the connecting rod.

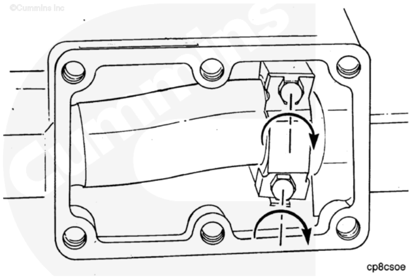
Make sure that the identification numbers on the connecting rod match.



Install new lock plates and the hexagon head capscrews.



Tighten the hexagon head capscrews.



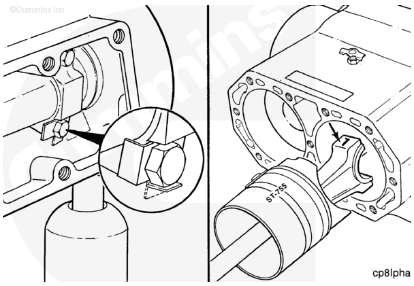
Torque Value:

1. 14 n•m [10 ft-lb]
2. 23 n•m [17 ft-lb]
3. Loosen both completely
4. 14 n•m [10 ft-lb]
5. 23 n•m [17 ft-lb]

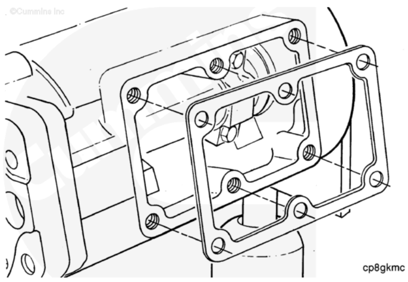
Rotate the crankshaft to make sure the rod journal is **not** binding on the crankshaft journal.

Use pliers to bend the lock plate tabs against the capscrew heads.

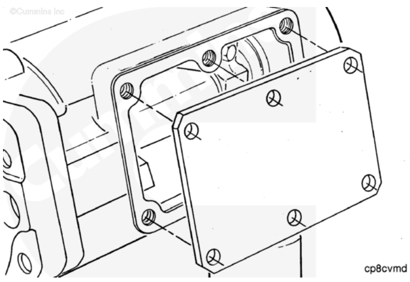
Repeat the last 19 steps to install the other piston and connecting rod assembly.



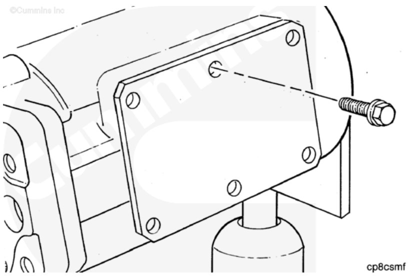
Install a new access cover gasket.



Install the access cover.

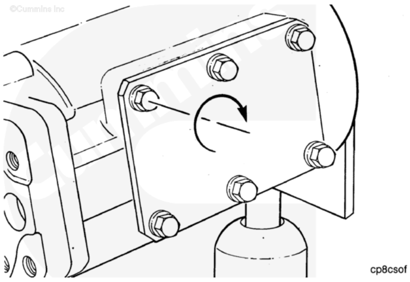


Install the three hexagon head capscrews and three flat head capscrews.

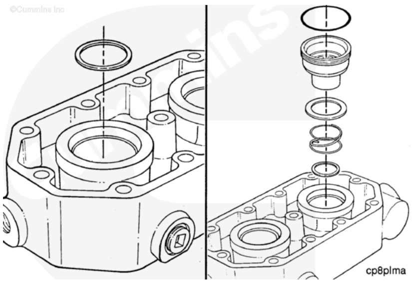


Tighten the capscrews.

Torque Value: 47 n•m [35 ft-lb]



Assemble the air compressor cylinder head. Refer to Procedure 012-106 in Section 12 (/qs3/pubsys2/xml/en/procedures/04/04-012-106.html).



Last Modified: 23-Apr-2019